

Simulations of Vortex Dynamics in Quantum Fluids

Alessio Zanella

17 October 2025

University of Verona

Introduction to Superfluidity

- The classical fluid dynamics has been known for more than a century, however, a complete understanding of the model (expecially under turbulent regimes) is still missing [3]
- In the context of quantum mechanics, the bosons can occupy simultaneously the same quantum state, leading to the formation of a superfluid phase [5], characterized by lack of viscosity and quantized vorticity
- The study of superfluid turbulence can help to understand classical turbulence, since the quantization of vorticity simplifies the dynamics

The Gross-Pitaevskii Equation

The governing equation for a weakly interacting Bose-Einstein condensate is the *Gross-Pitaevskii equation* [5]:

$$i\hbar \partial_t \psi = -\frac{\hbar^2}{2m} \nabla^2 \psi + V_0 |\psi|^2 \psi - E_0 \psi$$

where $\psi(\mathbf{r}, t)$ is the macroscopic wave function of the condensate, m is the mass of the bosons, V_0 is the interaction strength and E_0 is the chemical potential. It can be rewritten in dimensionless form [6] as:

$$\partial_t \psi = \frac{i}{2} \nabla^2 \psi + \frac{i}{2} (1 - |\psi|^2) \psi$$

Classical analogy via Madelung transformation i

To achieve a classical analogy, we can use the *Madelung transformation* [4]:

$$\psi(\mathbf{r}, t) = \rho^{1/2} \exp(iS) \quad (1)$$

where $\rho = \rho(\mathbf{r}, t) = |\psi(\mathbf{r}, t)|^2$ is the density and $S = S(\mathbf{r}, t)$ is the phase of ψ . The vector field $\mathbf{u} = \nabla S$ can be interpreted as the superfluid velocity field.

Substituting (1) into the GPE and separating real and imaginary parts, we obtain the following system, written in tensor notation:

Classical analogy via Madelung transformation ii

$$\frac{\partial \rho}{\partial t} + \frac{\partial \rho u_j}{\partial x_j} = 0$$
$$\rho \left(\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j} \right) = -\frac{\partial p}{\partial x_i} + \frac{\partial \tau_{ij}}{\partial x_j}, \quad i = 1, 2, 3$$

We can observe that this system is very similar to the Navier-Stokes equations, except for the pressure and stress term. This to justify the introduction.

Approximation of steady-state vortex i

In a two dimensional domain, we set $\psi(x, y) = f(r) \exp(i\theta(x, y))$ $r = \sqrt{x^2 + y^2}$, where we imposed the time independency. In order to obtain an initial approximaion of the straight vortex, we substitute the latter equation into GPE, imposing the time independency. We obtain the boundary value problem

$$f'' + \frac{f'}{r} + f \left(1 - f^2 - \frac{1}{r^2} \right) = 0$$

with boundary values $f(0) = 0, f(\infty) = 1$.

One can integrate this equation numerically, however, it is convenient to approximate f with a Padé approximation [1] in order to evaluate the latter

Approximation of steady-state vortex ii

everywhere. The approximant used in what follows is a 4-th order diagonal Padé approximation, which coefficients are reported in [2].

Space discretization i

- The central finite differences over uniform grid is simple and easy-to-implement
- The solution has a stepped gradient near the vortex core
- In order to achieve high resolution near the core, it is mandatory to use a large number of points, especially when considering large domains
- A solution to this issue could be employing a non uniform space discretization, clustering points near the vortex core.

Space discretization ii

Consider the interval $[a, b] \subset \mathbb{R}$, and let $\mathbf{x} = [x_1, \dots, x_m]^T$, with $x_1 = a$, $x_m = b$ and $x_j = x_{j-1} + h_j$, $1 \leq j \leq m-1$, where h_j satisfy the recurrence $h_j = (1 + \delta)h_{j-1}$ for some fixed parameter $\delta > 0$.

We can see that $h_j = (1 + \delta)h_{j-1} = (1 + \delta)^2 * h_{j-2} = \dots = (1 + \delta)^{j-1}h_1$.

Since we must cover the entire interval length, the constraint reads

$$\sum_{j=1}^{m-1} h_j = h_1 \sum_{j=1}^{m-1} (1 + \delta)^{j-1} = b - a$$

Space discretization iii

In order to compute the points, we can proceed in three natural ways, each consisting in fixing two parameters among m, δ and h_1 , then computing the remaining one.

The way that gives the most control on the grid construction is fixing δ and h_1 , and compute m as follows:

$$m = \left\lceil \frac{\log\left(\frac{\delta(b-a)}{h_1} + 1\right)}{\log(1 + \delta)} + 1 \right\rceil$$

The ceiling function is given by the fact that m must be an integer.

References i

-  Natalia G. Berloff.
Padé approximations of solitary wave solutions of the gross-pitaevskii equation.
Journal of Physics A: Mathematical and General, 37(5):1617 – 1632, 2004.
Cited by: 91; All Open Access, Green Open Access.
-  M. Caliari and S. Zuccher.
Reliability of the time splitting fourier method for singular solutions in quantum fluids.
Computer Physics Communications, 222:46–58, 2018.
See also arXiv:1603.05022.

References ii



Uriel Frisch.

Turbulence: The Legacy of A. N. Kolmogorov.

Cambridge University Press, 1995.



Erwin Madelung.

Quantentheorie in hydrodynamischer form.

Zeitschrift für Physik, 40:322–326, 1927.



Lev P. Pitaevskii and Sandro Stringari.

Bose–Einstein Condensation and Superfluidity.

Oxford University Press, 2016.



A.W. Baggaley S. Zuccher, M. Caliori and C.F. Barenghi.

Quantum vortex reconnections.

Physics of Fluids, 24(12):125108, 2012.